higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

FITTING AND TURNING
NQF LEVEL 3

11 MARCH 2019

This marking guideline consists of 6 pages.

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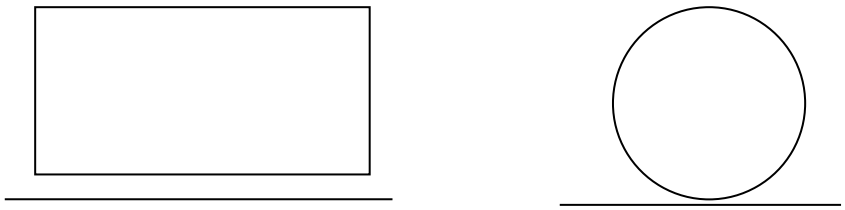
QUESTION 1: BEARINGS

1.1
- Workers or other people will not slip and injure themselves
- Core of good housekeeping practice
- Helps to work safely and efficiently (Any 2 × 1) (2)

1.2
- Contamination with dirt and foreign matter
- Abnormal load due to improper assembly
- Shaft and housing fits are too large or too small
- Improper lubricant or lubrication method
- Improper design
- Incorrect bearing installed (Any 5 × 1) (5)

1.3
- Induction-heater method
- Oil-bath method
- Heating-lamp method (3)

1.4



Line contact (roller bearing) Point contact (ball bearing) (2 + 2) (4)

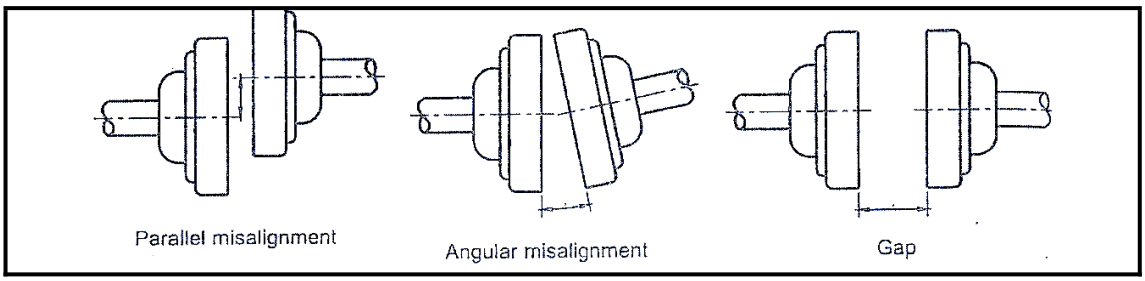
1.5 To keep rolling elements (balls or rollers) an equal distance apart around the circumference (1)

[15]

QUESTION 2: COUPLINGS

2.1
- Rigid/Permanent couplings
- Flexible couplings
- Self-aligning couplings (3)

2.2



(3 × 1) (3)

2.3
- Straight-edge and feeler-gauge method
- Rim-and-face method
- Reverse-indicator method
- Laser method (4)

[10]

QUESTION 3: BRAKES AND CLUTCHES

3.1
- The brake applies immediately if there is a power failure (good safety).
- It is a fast-response system. (2)

3.2
- Noise
- Shudder
- Slipping
- Overheating
- Nonengagement (Any 3 × 1) (3)

3.3
A – Outer cone
B – Lining
C – Inner cone
D – Key
E – Shaft (5)

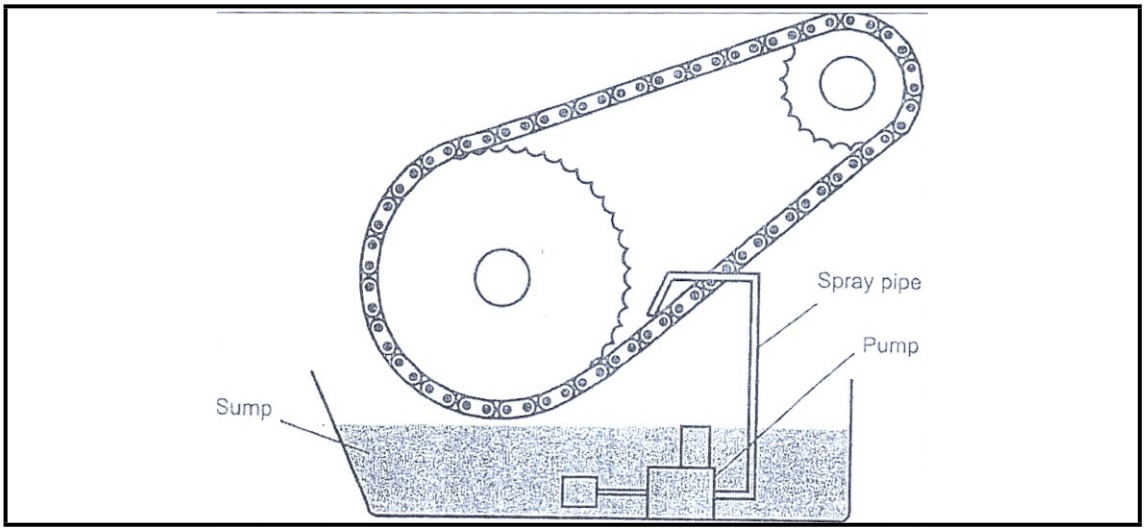
[10]

QUESTION 4: BELT DRIVES, CHAIN DRIVES AND GEAR DRIVES

4.1
- Transmits motion from one shaft to another
- Drives over a long distance (2)

4.2
- Prevents accidental contact to the drive
- Will not fly off and injure the worker when the belt breaks
- Prevents foreign material from entering the drive (3)

4.3



(Any relevant drawing)
(3 for labelling + 2 for drawing) (5)

4.4
- Positive drive (no slip)
- More compact
- More efficient
- High speed ratios
- More durable (Any 3 × 1) (3)

4.5
- Driven gear can rotate in the same direction as driver gear
- Varies the distance between the driver and driven gears (2)

[15]

QUESTION 5: PIPES, PIPE FITTINGS AND VALVES

5.1
- Bolts and nuts are not tight enough.
- Flanges do not fit properly to each other.
- Sealing devices are damaged.
- There is corrosion and cracks on the flanges.
- Damage occurs as a result of vibrations.
- Flanges are not pulled up evenly. (Any 5 × 1) (5)

5.2
- Always wear safety clothing (PPE).
- Appropriate safety guards must be in place.
- Isolate the valve before inspections and servicing.
- Depressurise the system.
- All tools and equipment should be on hand when doing repairs or maintenance. (Any relevant 5 × 1) (5)

[10]

QUESTION 6: CENTRE LATHES

6.1
a Headstock
b Carriage
c Tailstock
d Tool post
e Compound slide
f Cross slide
g Apron
h Feed shaft
i Lead screw

(9 × 1) (9)

6.2 D = 10 mm = 10/1 000 = 0,01 m
N = 1 800 r/min

$S = \pi \times D \times N$
$= \pi \times 0,01 \times 1\,800$
$= 56,549$ m/min (3)

6.3
- Finish required
- Type of material
- Type of tool used
- Diameter of workpiece (4)

6.4
- Facing
- Parallel turning (longitudinal)
- Grooving or parting off
- Taper turning
- Drilling or boring
- Thread cutting (Any 3 × 1) (3)

6.5 Shields or guards prevent chips from flying to the operator which can cause injuries and also prevent the chips from flying all over onto the floor. (1)
[20]

QUESTION 7: MILLING MACHINES

7.1 7.1.1 A roughing cutter is used to remove large amounts of material.

7.1.2 An end mill is used for side cutting, slots and flat surfaces.

7.1.3 A slot drill is used for plunging operations such as drilling vertically downwards for a certain depth, milling horizontally and also machining a keyway.

7.1.4 A T-slot is used for making T-slots in machine tables.

7.1.5 A dovetail cutter is used for cutting internal or external dovetails.
(5 × 1) (5)

7.2
- Wear goggles/Safety glasses.
- Clamp the workpiece securely.
- Never leave the machine unattended.
- Make sure the cutting tool is secured.
- Do not make any adjustments while the machine is running. (Any 3 × 1) (3)

7.3
- Rapid (direct) indexing
- Simple indexing
- Angular indexing
- Differential indexing (4)

7.4 $S = \pi \times D \times N$

$N = S/\pi \times D$
$= 10/ \pi \times 0,02$
$= 159,2$ r/min

$f = f_t \times T \times N$
$= 0,04 \times 4 \times 159,2$
$= 25,5$ mm/min (6)

7.5
- Solid collet
- Split collet (2)
[20]

TOTAL: 100

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